

Linear Regression, and Model Assessment/Selection

Week 2

Review from Last Week

Machine learning is the science of learning patterns from data.

We introduced three types of machine learning problems:

- **Supervised Learning**
- **Unsupervised Learning**
- Reinforcement Learning

The main focus of this course is supervised learning, where we are given

- **Training Set** $(x_1, y_1), \dots, (x_n, y_n)$ consisting of n samples of
 - **features** $x_1, \dots, x_n \in X$ and corresponding
 - **outcome** $y_1, \dots, y_n \in Y$

and the goal is to estimate (or learn) the function $f : X \rightarrow Y$ based on D^{train} .

Review from Last Week

We discussed two approaches of estimating f based on D^{train} :

- **Parametric method** (assume a distribution)
 - *Example*: assuming a linear model
- **Non-parametric method** (data-driven)
 - *Example*: k-nearest neighbours

When choosing a model, $\hat{f}$, we need to keep in mind the **bias-variance tradeoff**

- the balance between a model being too simple (high bias) and too complex (high variance)

Review from Last Week

When deciding on the best model, we can use a metric such as the **test MSE** (regression problems) or the **test error rate** (classification problems)

- it is important to evaluate model performance using the test data, since the training MSE (or error rate) will always prefer a more complex model

In summary, different $\hat{f}$ are simply different algorithms/methods/predictors

- we will spend this course going over many different ways to estimate f
- there is no one best method

Learning Outcomes for This Week

By the end of this week, you will be able to...

- Explain and apply ordinary least squares (OLS) for fitting a linear regression model and interpret the resulting coefficient estimates.
- Explain the need for model assessment and selection, including the distinction between training error and expected test error.
- Apply validation-set, leave-one-out, and k -fold cross-validation to estimate test error and compare competing models.
- Compare common model-selection criteria, including Mallows' C_p , adjusted R^2 , AIC, and BIC.
- Describe and compare subset-selection methods, including best subset, forward stepwise, and backward stepwise selection, and discuss their computational trade-offs.

Linear Regression: A Review

Motivating Example

Chronic obstructive pulmonary disease (COPD) is a long-term lung condition that makes it progressively harder to breathe.

Despite the decreasing rates in cigarette smoking, COPD continues to be a problem.

We wish to:

- 1. better understand the features that are associated with COPD, and
- 2. predict COPD in new patients.

Since we care about the interpretability of the model, a natural choice is a linear model to estimate f !

Kamis, A., Gadia, N., Luo, Z., Ng, S. X., & Thumbbar, M. (2024). Obtaining the Most Accurate, Explainable Model for Predicting Chronic Obstructive Pulmonary Disease: Triangulation of Multiple Linear Regression and Machine Learning Methods. JMIR AI, 3, e58455.

Linear Regression

Although simple, and may seem a bit dull compared to other more sophisticated techniques, linear regression remains to be a very useful and widely used statistical learning method.

It is the foundation of a lot of new approaches as well.

Linear Regression

Let $Y \in \mathbb{R}$ be the outcome and $X \in \mathbb{R}^p$ be the (random) vector of p features.
The linear model assumes:

$$\begin{aligned} Y &= f(X) + \epsilon \\ &= \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon \end{aligned}$$

where

- $\beta_0, \beta_1, \dots, \beta_p$ are unknown constants
- β_0 is called the **intercept**
- β_j , for $1 \leq j \leq p$, are the **coefficients** of the p features
- ϵ is the error term satisfying $\mathbb{E}[\epsilon] = 0$

Linear Predictor under the Linear Regression Model

If we can estimate the unknown coefficients $\beta_0, \dots, \beta_p$, then we can use the fitted model to predict the response for a new observation $\mathbf{x}^* = (x_1^*, \dots, x_p^*)$:

$$\hat{f}(\mathbf{x}^*) = \hat{\beta}_0 + \hat{\beta}_1 x_1^* + \dots + \hat{\beta}_p x_p^*.$$

Question: How do we estimate $\hat{\beta}_0, \dots, \hat{\beta}_p$?

Ordinary Least Squares (OLS) Approach

To approximate $\beta_0, \dots, \beta_p$, we can use Ordinary Least Squares (OLS).

In the **model fitting step**, we use the training data, D^{train} to choose $\beta_0, \dots, \beta_p$ such that they minimize the mean of the squared residuals:

$$(\hat{\beta}_0, \dots, \hat{\beta}_p) = \arg \min_{\alpha_0, \dots, \alpha_p} \frac{1}{n} \sum_{i=1}^n (y_i - [\alpha_0 + \alpha_1 x_{i1} + \dots + \alpha_p x_{ip}])^2.$$

Once we have estimated the coefficients, our fitted model is

$$\hat{f}(\mathbf{x}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p.$$

Linear Regression

Question: What is the interpretation of β_0 ?

the outcome when all predictors have a value of zero

Question: What is the interpretation of one of the coefficients, say, β_1 ?

for a one unit increase in X_1 , the outcome will increase (or decrease, depending on the sign) by β_1 , keeping all other predictors constant

Motivating Example

Multiple linear regression.	
	Estimate
(Intercept)	32.4000
Smoking_Rate	0.2570
Log_HH_Income	−2.8100
Hispanic_percentage	−2.3900
Education_Rate	−0.0334
AI_or_AN_percentage ^a	−2.9000
Black_percentage	−1.2700
NH_or_PI_percentage ^b	−15.8000
Log_GDP	0.0741
White_percentage	−0.7380
Unemployment_Rate	0.0456
Asian_percentage	2.4800
Log_Population	0.0899
Air_Quality_Index	0.0003

In the analysis of the paper our motivating example is based on, the **outcome** is the COPD rate in the population for a specific region.

They used the **features** on the left to define their linear model, and estimated the coefficients using OLS.

Question: What is the interpretation of the coefficient associated with the smoking rate of the region?

In Matrix Form

Using matrix notation,

- $\hat{\boldsymbol{\beta}} = [\hat{\beta}_0, \dots, \hat{\beta}_p]^\top \in \mathbb{R}^{p+1}$, $\boldsymbol{\alpha} = [\alpha_0, \dots, \alpha_p]^\top \in \mathbb{R}^{p+1}$,
- $\mathbf{y} = [y_1, \dots, y_n]^\top \in \mathbb{R}^n$, $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top \in \mathbb{R}^{n \times (p+1)}$, with

$$\mathbf{x}_i = [1, x_{i1}, \dots, x_{ip}]^\top \in \mathbb{R}^{p+1}, \quad 1 \leq i \leq n.$$

The OLS estimator of $\boldsymbol{\beta}$ is defined as

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \arg \min_{\boldsymbol{\alpha} \in \mathbb{R}^{p+1}} \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\alpha})^2 \\ &= \arg \min_{\boldsymbol{\alpha} \in \mathbb{R}^{p+1}} \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}\|_2^2. \end{aligned}$$

Some Comments

- We use the **test MSE** to evaluate the performance of a model, but the idea of estimating f by **minimizing the training MSE** can be applied to (almost) all supervised learning problems.

Specifically,

$$\hat{f} = \arg \min_{g \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(y_i, g(\mathbf{x}_i))$$

where $L(\cdot)$ is a loss function and $\mathcal{F}$ is a class of choices for the function, g .

- The OLS approach corresponds to $L(a, b) = (a - b)^2$ and

$$\mathcal{F} = \{\mathbf{x} \mapsto \mathbf{x}^\top \beta : \beta \in \mathbb{R}^{p+1}\}$$

Some Comments

- In general, the difficulty of solving

$$\hat{f} = \arg \min_{g \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(y_i, g(\mathbf{x}_i))$$

varies across problems, but the OLS approach admits a closed-form solution:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

whenever $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ has full column rank.

What is Random?

The true parameters β are **fixed but unknown**.

The training data

$$(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)$$

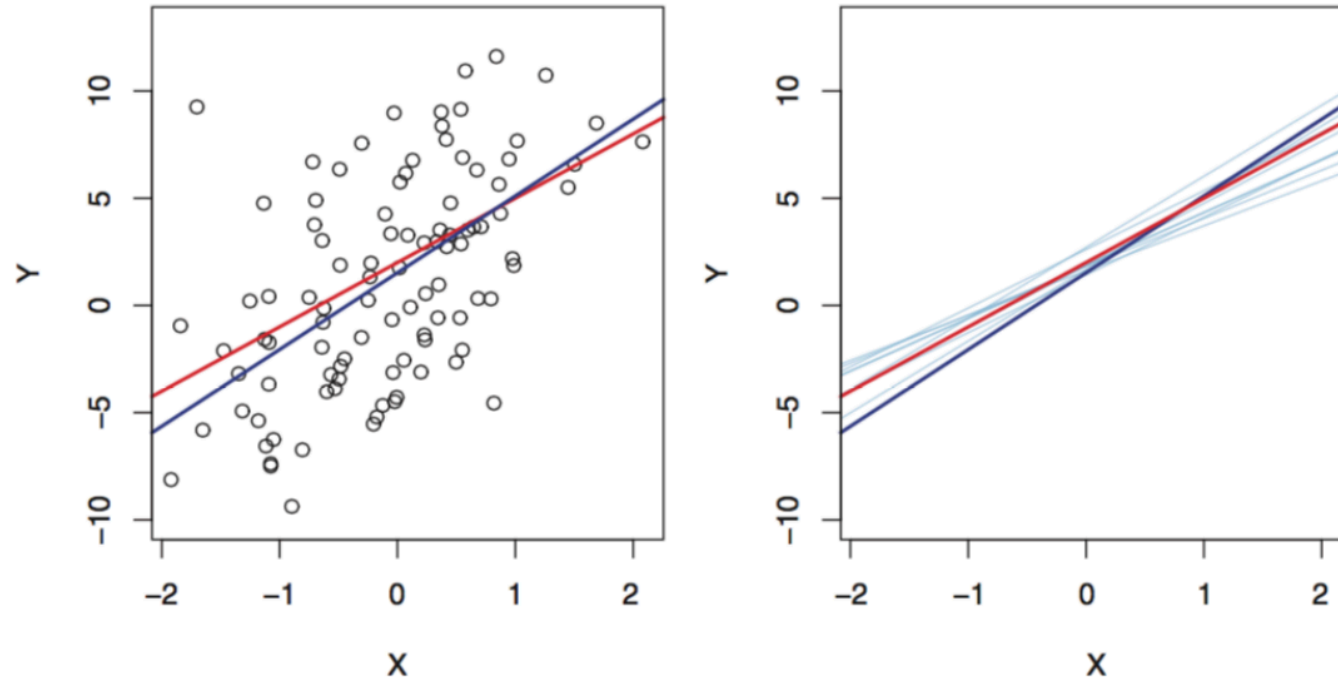
are random observations from the population.

Since $\hat{\beta}$ is calculated from the training data,

$\hat{\beta}$ is also random.

If we collected a different training sample, we would generally obtain a **different** $\hat{\beta}$.

The Randomness in the Coefficients



Left: The **red line** represents the true model $f(X) = 2 + 3X$. The **blue line** is the OLS fit based on the observed data.

Right: The light blue lines represent 10 OLS fits, each one computed on the basis of a different training dataset.

The fitted OLS lines are different, but their average is close to the true regression line.

Properties of the OLS Estimator

Recall that

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Assuming the linear model is correctly specified,

- **Unbiasedness**: $\mathbb{E}[\hat{\beta}] = \beta$.
- The **covariance matrix** of $\hat{\beta}$ is:

$$\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Together, these determine the **expected squared estimation error** (ℓ_2):

$$E \left[\|\hat{\beta} - \beta\|_2^2 \right] = \sigma^2 \text{trace} \left[(\mathbf{X}^\top \mathbf{X})^{-1} \right].$$

What Affects Estimation Error?

Consider the simple case where the predictors are orthogonal and standardized:

$$X^{\top} X = nI_{p+1}.$$

In this case, the ℓ_2 estimation error is:

$$E \left[\|\hat{\beta} - \beta\|_2^2 \right] = \frac{\sigma^2(p+1)}{n}.$$

This tells us that estimation error:

- $\uparrow$ as the **noise σ^2 increases**
- $\uparrow$ as the **number of predictors p increases**
- $\downarrow$ as the **sample size n increases**

Even though OLS is unbiased, its estimates can have high variance.

Important Considerations

Estimation of β

- How close is the point estimation $\hat{\beta}$ to the true β
- (Inference) Can we provide confidence intervals and conduct hypothesis testing of $\hat{\beta}$?

Prediction of Y at $X^* = \mathbf{x}^*$

- How accurate is the point prediction $\hat{f}(\mathbf{x}^*) = \mathbf{x}^{*\top} \hat{\beta}$?
- (Inference) Can we further provide prediction intervals of Y ?

Variable (Model) Selection

- Do all predictors help to explain Y , or is there only a subset of the predictors that are useful? (we will discuss this later!)

Summary

- Linear regression, although simple, remains a very popular model because of its simplicity and interpretability
- When fitting a linear model to training data, the problem is reduced to estimating the coefficients, $\beta_i, i = 0, \dots, p$
- We can use ordinary least squares (OLS) to do this estimation
- the OLS approach involves minimizing the **training MSE** to estimate $\hat{f}$
- OLS estimates of the coefficients are unbiased
- Once we have a linear model, how do we assess the performance? And how can we select models that better fit the data?

Model Assessment

How Do We Choose Among Models?

Suppose we are considering two models:

$$\text{Model 1:} \quad Y = \alpha_0 + \alpha_1 X_1 + \epsilon,$$

$$\text{Model 2:} \quad Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

These lead to different predictors at $\mathbf{X} = \mathbf{x} = (x_1, x_2)$:

$$\hat{f}_1(\mathbf{x}) = \hat{\alpha}_0 + \hat{\alpha}_1 x_1$$

versus

$$\hat{f}_2(\mathbf{x}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2.$$

Ideally, we choose the model with the **smaller expected test MSE**.

How Do We Compare Expected MSEs?

If we have an independent test set, D^{test} , we can compare the test MSEs directly.
For Model 1:

$$\frac{1}{m} \sum_{i=1}^m \left(y_i^{(T)} - \hat{\alpha}_0 - \hat{\alpha}_1 x_{i1}^{(T)} \right)^2$$

and for Model 2:

$$\frac{1}{m} \sum_{i=1}^m \left(y_i^{(T)} - \hat{\beta}_0 - \hat{\beta}_1 x_{i1}^{(T)} - \hat{\beta}_2 x_{i2}^{(T)} \right)^2.$$

Question: What if we don't have an independent test set?

Model Assessment

When we do not have a test set, there are two common approaches to model assessment.

Estimate the expected test error

We can manually create a “test set” using **data-splitting techniques**:

Adjust the training error

Alternatively, we can adjust the training error to account for **model complexity**:

Direct Estimation of the Expected MSE: Data-Splitting Techniques

One approach is to **randomly split** the available data to create a validation set that functions as a test set.

Validation set approach:

One-time data splitting.

Cross-validation approach:

Multiple data splits.

Validation Set Approach

Randomly divide the available observations into two parts:

- a **training set**
- a **validation (hold-out) set**

The proportion assigned to each set depends on the application.

The model is **fitted on the training set** and **evaluated on the validation set**.

The resulting validation-set MSE provides an estimate of the expected test MSE.

Validation Set Approach

Consider a dataset that has n observations.

In the figure below, the 7th, 22nd, 13th observation are *randomly* assigned to the training set.

The 91st observation is *randomly* assigned to the validation set.

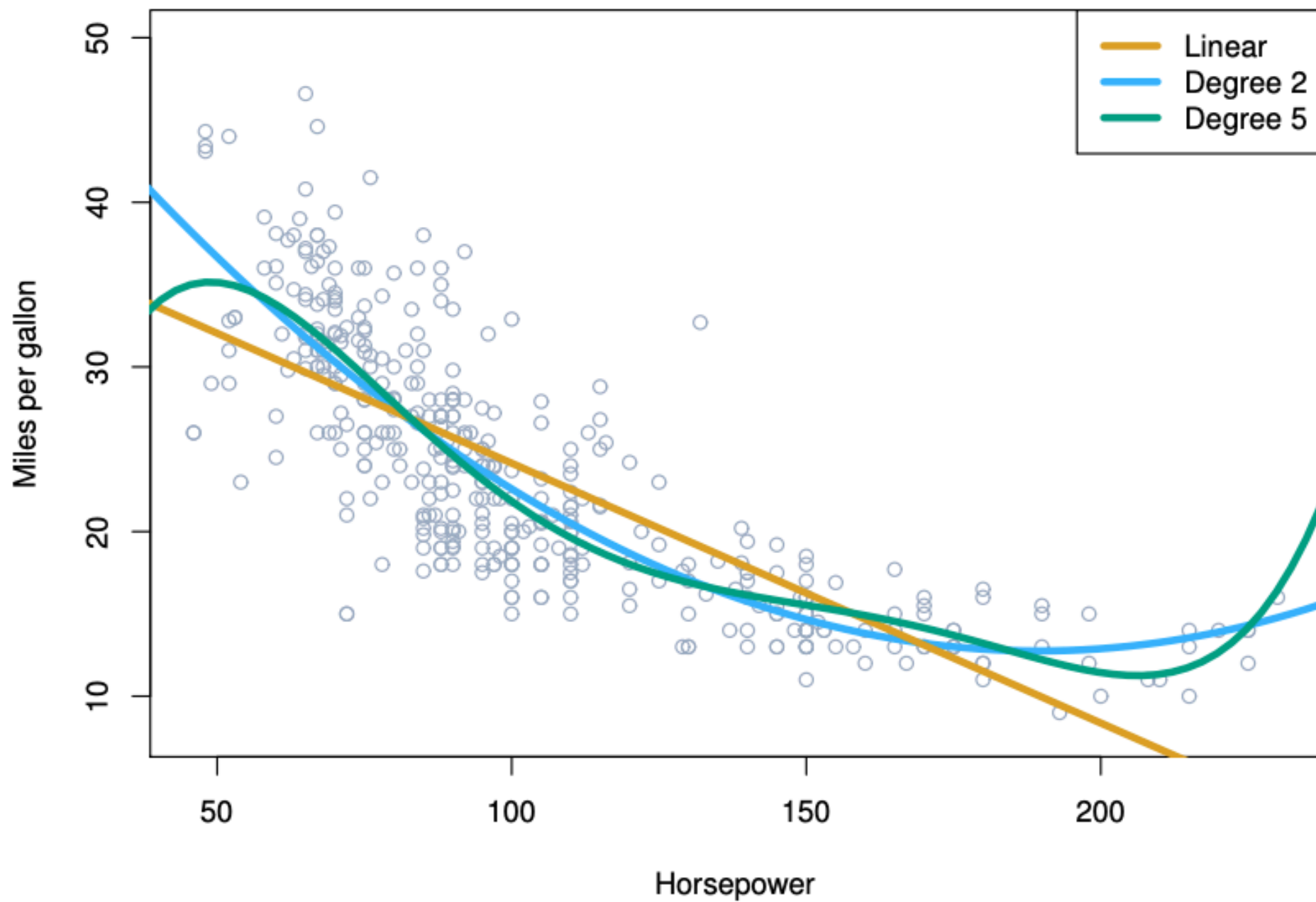


Example: Auto Data

This example is taken from the [ISL] textbook.

There is a **non-linear** relationship between mpg and horsepower.

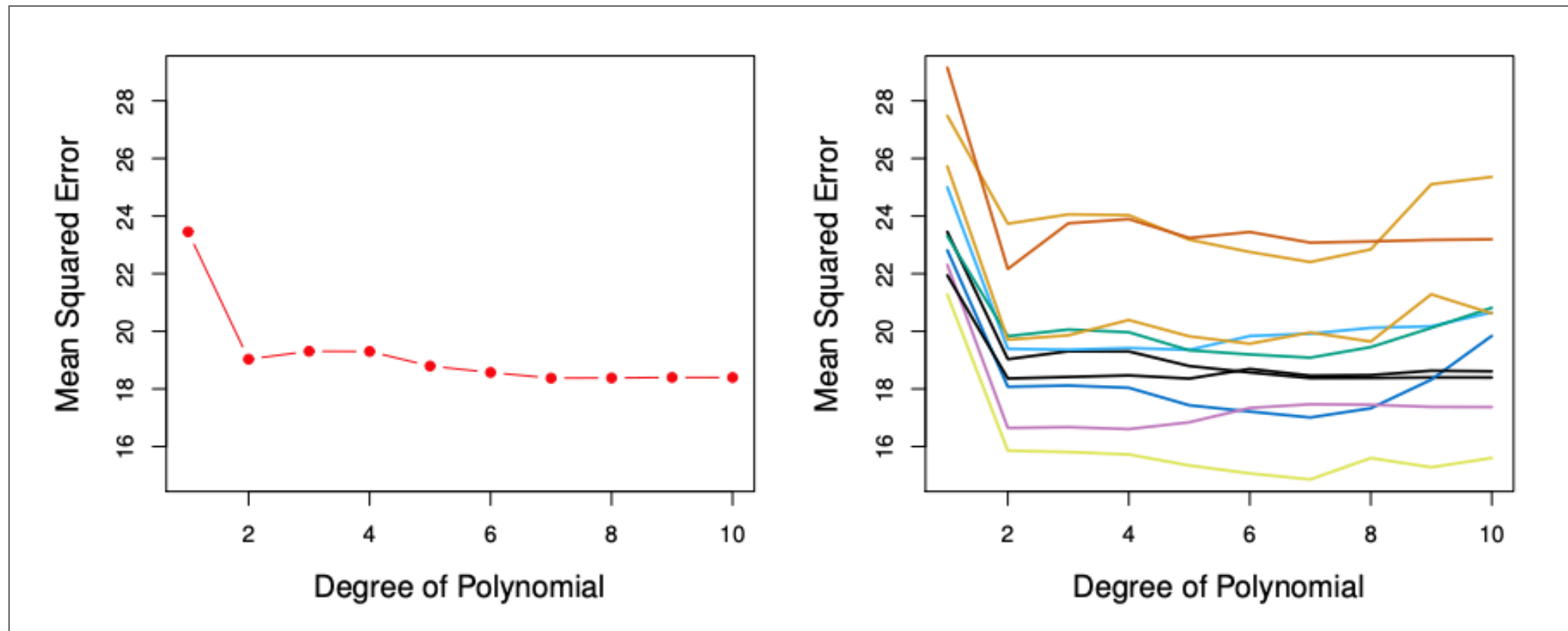
We might therefore consider comparing a linear model to other polynomials of higher degrees.



Example: Auto Data

We randomly split the **392 observations** into:

- 196 observations in the training set
- 196 observations in the validation set



Key observation: The estimated validation error can change substantially depending on the random split.

One-time data splitting is not very stable.

Drawbacks of the Validation Set Approach

The validation set approach has two important drawbacks.

The estimate can be highly variable.

The estimated test error depends on which observations happen to be placed in the training set and which are placed in the validation set.

We fit the model using only part of the available data.

Since fewer observations are used for training, the fitted model may perform worse than a model fit using the entire data set.

As a result, the validation-set error may tend to **overestimate** the test error of a model fit using all available observations.

Question: How can we remedy these drawbacks?

Leave-One-Out Cross-Validation (LOOCV)

Instead of creating one large validation set, suppose we leave out **only the first observation**.

The validation set is

$$(\mathbf{x}_1, y_1),$$

and the training set contains

$$(\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n).$$

Leave-One-Out Cross-Validation (LOOCV)

Using the training set, fit the model $\hat{f}_1$ and predict the response for the observation that was left out.

The corresponding squared error is

$$\text{MSE}_1 = \left(y_1 - \hat{f}_1(\mathbf{x}_1) \right)^2.$$

But using only **one** observation to evaluate our model is not enough!

Leave-One-Out Cross-Validation

Now repeat the procedure, leaving out the **second observation**.

The validation set is

$$(\mathbf{x}_2, y_2),$$

while the remaining $n - 1$ observations form the training set.

Fit $\hat{f}_2$ using the training data and calculate

$$\text{MSE}_2 = \left(y_2 - \hat{f}_2(\mathbf{x}_2) \right)^2.$$

Repeat this procedure n times, leaving out each observation once to obtain $\text{MSE}_1, \text{MSE}_2, \dots, \text{MSE}_n$.

The LOOCV estimate of the test MSE is the average of these n test MSE estimates:

$$\boxed{CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i}$$

Leave-One-Out Cross-Validation

In LOOCV, every observation serves as the validation observation **exactly once**.
The training set is shown in blue, the validation set in beige.

1 2 3 n



1 2 3 n

1 2 3 n

1 2 3 n

⋮

1 2 3 n

LOOCV vs. the Validation Set Approach

LOOCV has several advantages over the validation set approach.

More data are used for training.

Each LOOCV model is trained using $n - 1$ observations, so its training set is almost the entire data set.

There is no randomness in the split.

Unlike the validation set approach, repeating LOOCV on the same data produces the same result.

However, LOOCV can be **computationally expensive**, since the model generally needs to be fit n times.

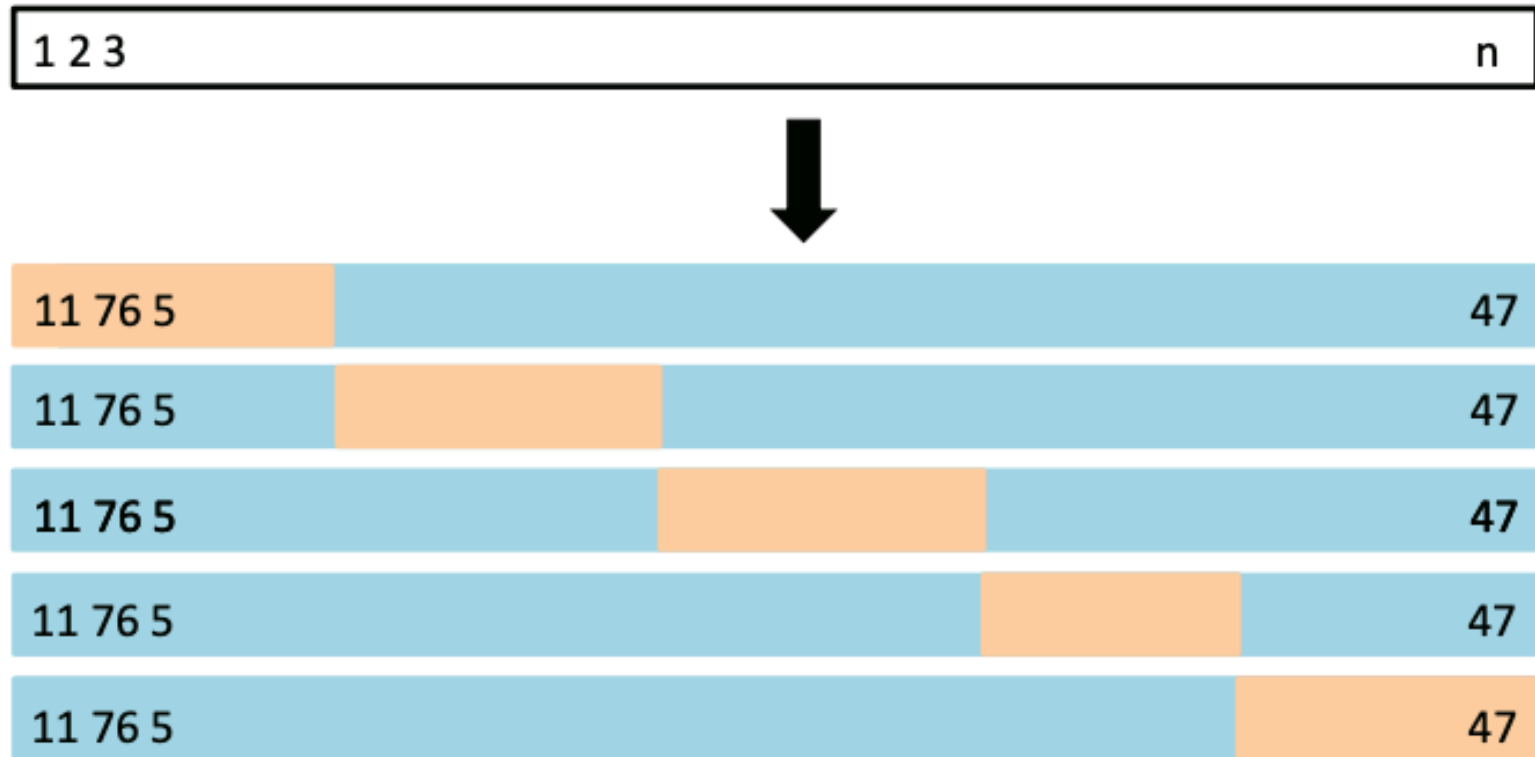
k -Fold Cross-Validation

Instead of leaving out one observation at a time, randomly divide the data into k approximately equal-sized groups called **folds**.

1. Treat the first fold as the validation set.
2. Fit the model using the remaining $k - 1$ folds.
3. Calculate the prediction error on the first fold, MSE_1 .
Repeat this procedure for folds $2, 3, \dots, k$ to obtain $\text{MSE}_1, \text{MSE}_2, \dots, \text{MSE}_k$.
The k -fold cross-validation estimate is

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k \text{MSE}_i$$

k -Fold Cross-Validation



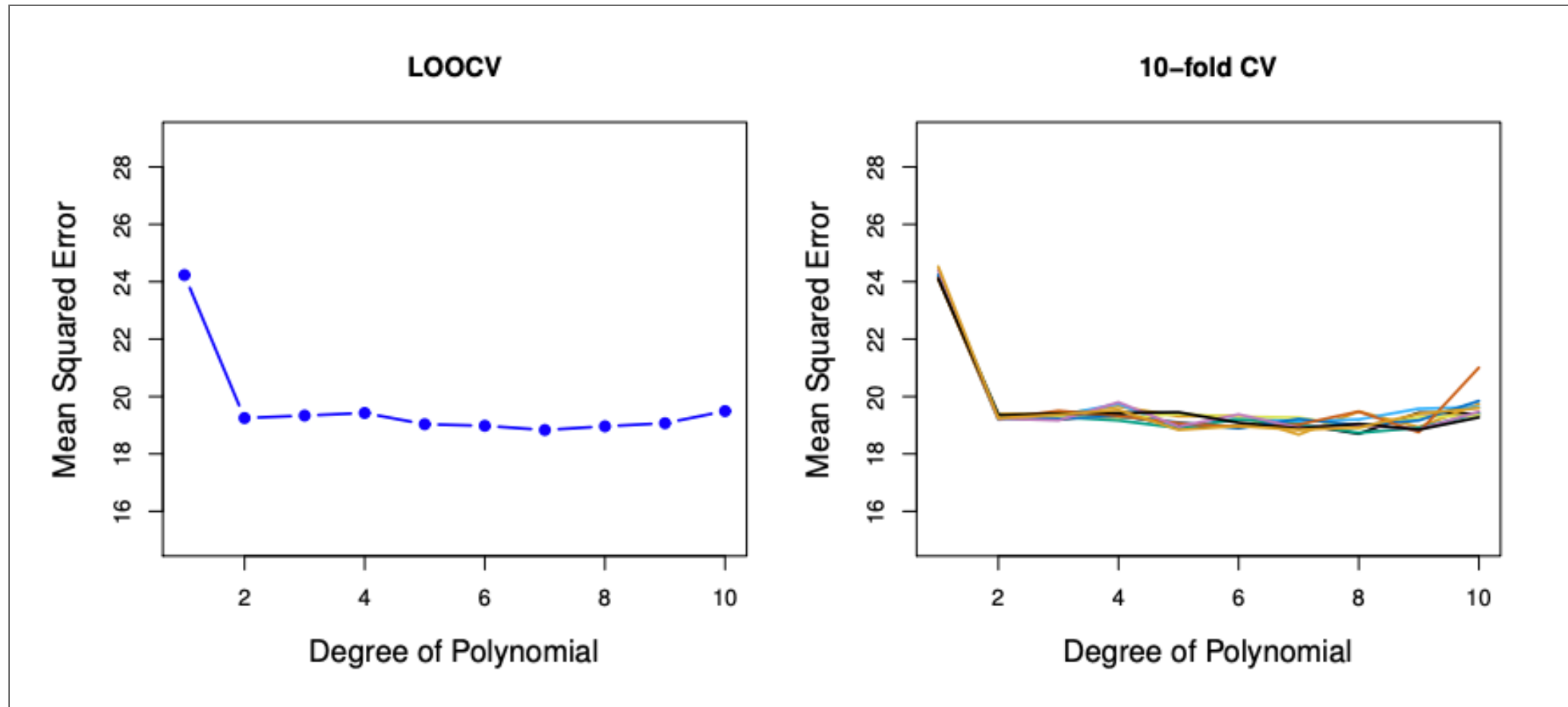
k -Fold Cross Validation

Some common choices are $k = 5$ or $k = 10$

LOOCV is a special case of k -fold cross-validation where

$$k = n.$$

Example: Auto Data



Left: LOOCV error curve

Right: 10-fold cross validation was run nine separate times each with a different random split.

Cross-Validation for Classification

We have focused on regression problems, but cross-validation can also be used for **classification problems**.

For LOOCV, we split the data in exactly the same way as before.

Instead of squared error, we can calculate whether the validation observation was classified correctly:

$$\text{Err}_i = \mathbb{1} \left\{ y_i \neq \hat{f}_i(\mathbf{x}_i) \right\}.$$

Thus,

$$\text{Err}_i = \begin{cases} 1, & \text{if the observation is misclassified,} \\ 0, & \text{if the observation is classified correctly.} \end{cases}$$

The LOOCV estimate of the classification error is

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{Err}_i$$

Cross-Validation Can Be Tricky!

Key idea: Independence between the fitted model and the validation set is essential.

The validation observations should **not be used to fit the model**.

A Tricky Example

Suppose we have:

- $n = 100$ observations
- $p = 5000$ predictors

We use the following two-step procedure:

1. Find the **10 predictors with the largest correlation with the outcome**.
2. Fit OLS using only these 10 predictors.

Question:

How should we use cross-validation to estimate the expected test MSE of this entire procedure?

Knowledge Check

1. Suppose we use 10-fold cross-validation. How many times is each observation used for **validation**? How many times is it used for **training**?

one time for validation, nine times for training

2. Suppose two models have training MSEs of 4.2 and 2.1, but their cross-validation MSEs are 5.0 and 7.3, respectively. Which model would you choose? What might explain these results?

first model; since second model has such a large difference between the training and testing MSE this suggests overfitting

3. Suppose we increase k in k -fold cross-validation from $k = 5$ to $k = 10$. What happens to the size of the training set used in each fit? What happens to the size of each validation set?

training set increases in size, validation set decreases in size

Avoid Estimating the Expected MSE: Adjust the Training Error

Instead of directly estimating the expected test MSE using data splitting, we can adjust the **training error** to account for model complexity.

Common approaches include:

- Mallows' C_p
- Adjusted R^2
- AIC
- BIC

These approaches are generally used for **parametric models**, such as linear regression, and typically rely on assumptions about the model and likelihood.

Recall: Training Error

For a fitted model $\hat{f}$, let $\hat{f}(\mathbf{x}_i)$ denote the fitted value for observation i .
For a linear regression model with p predictors,

$$\hat{f}(\mathbf{x}_i) = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}} = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}.$$

The residual for observation i is $y_i - \hat{f}(\mathbf{x}_i)$.

The residual sum of squares is

$$RSS(\hat{f}) = \sum_{i=1}^n \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2.$$

Therefore, $\frac{1}{n} RSS(\hat{f})$

is simply the **training MSE**.

Important: Training error generally **decreases as model complexity increases**.

Mallows' C_p

Let p denote the number of parameters in the model. Then, Mallows' C_p is defined as

$$C_p(\hat{f}) = \frac{1}{n}RSS(\hat{f}) + \frac{2p\sigma^2}{n}.$$

When σ^2 is unknown, we replace it with an estimate $\hat{\sigma}^2$.

The first term rewards **good fit**:

$$\frac{1}{n}RSS(\hat{f}).$$

The second term penalizes **model complexity**:

$$\frac{2p\hat{\sigma}^2}{n}.$$

We select the model with the **smallest** C_p .

C_p is primarily used for selecting among linear regression models.

$$R^2$$

Recall that the total sum of squares is

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2,$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.

Then, recall that

$$R^2(\hat{f}) = 1 - \frac{RSS(\hat{f})}{TSS}.$$

Because RSS can only decrease when predictors are added, R^2 can only **increase** when predictors are added to the model.

Adjusted R^2

Adjusted R^2 accounts for the number of predictors:

$$R_{\text{adj}}^2(\hat{f}) = 1 - \frac{RSS(\hat{f})/(n - p - 1)}{TSS/(n - 1)}.$$

Unlike ordinary R^2 , adjusted R^2 penalizes the inclusion of unnecessary predictors.

While $RSS(\hat{f})$ always decreases as p increases, $\frac{RSS(\hat{f})}{n-p-1}$ may increase or decrease.

We prefer models with **larger adjusted R^2** .

Both Mallows' C_p and adjusted R^2 are primarily used for selecting among **linear regression models**.

AIC and BIC

Suppose $\hat{f}$ is obtained by maximum likelihood estimation, and let $L(\hat{f})$ denote the maximized likelihood.

The **Akaike Information Criterion (AIC)** is

$$AIC(\hat{f}) = -2 \log L(\hat{f}) + 2p$$

Note: In the linear model with $\epsilon_i \sim N(0, \sigma^2)$, $AIC(\hat{f})$ is proportional to $C_p(\hat{f})$, selecting the same model.

The **Bayesian Information Criterion (BIC)** is

$$BIC(\hat{f}) = -2 \log L(\hat{f}) + (\log n)p$$

Note: BIC places a heavier penalty on models with many predictors, and hence results in selecting smaller-size models compared to the AIC and C_p .

AIC vs. BIC

Both criteria balance:

model fit vs. model complexity.

For both AIC and BIC, we select the model with the **smallest value**.

To compute AIC and BIC, we need to specify the likelihood, i.e. the distribution of $Y|X$, and to compute the maximum likelihood estimator

AIC and BIC can also be used for selecting parametric models in classification problems.

AIC and BIC

To calculate AIC or BIC, we must specify a likelihood, i.e. a probability model for

$$Y \mid X.$$

We then fit the model using maximum likelihood.

Unlike C_p and adjusted R^2 , AIC and BIC can also be used for other parametric models, including **classification models** such as logistic regression.

Example

Suppose we are comparing

$$\text{Model 1:} \quad Y = \alpha_0 + \alpha_1 X_1 + \epsilon,$$

$$\text{Model 2:} \quad Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

Their fitted predictors are

$$\hat{f}_1(\mathbf{x}) = \hat{\alpha}_0 + \hat{\alpha}_1 x_1 \quad \text{and} \quad \hat{f}_2(\mathbf{x}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2.$$

Their residual sums of squares are

$$RSS(\hat{f}_1) = \sum_{i=1}^n (y_i - \hat{\alpha}_0 - \hat{\alpha}_1 x_{i1})^2$$

and

$$RSS(\hat{f}_2) = \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} \right)^2.$$

Example

For Model 1,

$$C_p(\hat{f}_1) = \frac{1}{n} \left[RSS(\hat{f}_1) + (2 \times 1)\sigma^2 \right].$$

For Model 2,

$$C_p(\hat{f}_2) = \frac{1}{n} \left[RSS(\hat{f}_2) + (2 \times 2)\sigma^2 \right].$$

Similarly,

$$R_{\text{adj}}^2(\hat{f}_1) = 1 - \frac{RSS(\hat{f}_1)/(n-2)}{TSS/(n-1)}$$

and

$$R^2_{\text{adj}}(\hat{f}_2) = 1 - \frac{RSS(\hat{f}_2)/(n-3)}{TSS/(n-1)}.$$

Data Splitting vs. Complexity Criteria

Data-splitting methods have several advantages over criteria such as C_p , adjusted R^2 , AIC, and BIC.

- They provide a **direct estimate of test error**.
- They can be used for a wide range of models.
- They do not require us to explicitly determine model degrees of freedom.
- They do not always require an estimate of the error variance.

However, data splitting also has drawbacks:

- It can require a relatively large sample size.
- Cross-validation can be computationally expensive.
- Theoretical guarantees for the selected model may be difficult to obtain.

Knowledge Check

1. Suppose we fit five nested linear regression models, where each model adds one additional predictor. The training RSS decreases for every model.

Should we conclude that the model with the smallest RSS is the best model?

no, RSS will always decrease as we add predictors; better to use adjusted RSS which penalizes the addition of unnecessary predictors

2. Suppose we want to choose between several k -Nearest Neighbours models.

Which of the model-selection approaches we have discussed would you use?

cross validation since kNN is non parametric (the other methods need distributional information)

What's Next?

We now have tools for comparing models:

- Cross-validation
- Mallows' C_p
- Adjusted R^2
- AIC
- BIC

Next, we apply these tools to **model selection under the linear model**.

Model Selection

Review

To estimate the coefficients related to linear regression (in order to estimate f), we covered the OLS approach:

$$\hat{f}(\mathbf{x}) = \mathbf{x}^\top \hat{\boldsymbol{\beta}},$$

where

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\alpha}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}\|_2^2.$$

We have also learned some statistical properties of $\hat{\boldsymbol{\beta}}$ under the linear model

$$Y = \beta_0 + X_1\beta_1 + \cdots + X_p\beta_p + \epsilon.$$

- Unbiasedness
- Estimation error (ℓ_2 -norm)

Why Consider Alternatives to OLS?

There are two main motivations for considering alternative fitting procedures to OLS.

Prediction / Estimation

The OLS estimator

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

can have large variance when p is large.

In particular, if $p > n$, the OLS estimator is not unique and its variance is infinite.

Interpretability

By removing irrelevant features – that is, by setting some coefficient estimates to zero – we can obtain a more **parsimonious** and interpretable model.

What Are the Alternatives?

1. **Subset Selection**: Identify a subset of the p predictors that we believe are related to the response, and fit a model using OLS on the selected predictors.
 - Best subset selection
 - Stepwise selection
2. **Shrinkage**: Fit a model involving all p predictors, but shrink the estimated coefficients towards zero relative to the OLS estimates.

This is also known as **regularization** and can reduce variance. Some shrinkage methods can also perform variable selection.

 - Ridge
 - Lasso
3. **Dimension Reduction**: Project the p predictors into an r -dimensional subspace, where $r < p$, and use the resulting r projections as new predictors in a linear regression model.
 - We will discuss later in the term

Subset Selection

Best Subset Selection

Suppose we have three predictors:

$$\mathbf{X} = (X_1, X_2, X_3).$$

Question:

If we want to fit a linear regression model, what are **all possible subsets** of predictors?

$\{0\}, \{X_1\}, \{X_2\}, \{X_3\}, \{X_1, X_2\}, \dots, \{X_1, X_2, X_3\}$

Best Subset Selection

Algorithm: Best Subset Selection

1. Let $\mathcal{M}_0$ denote the null model containing no predictors.
 2. For $k = 1, \dots, p$:
 - a. Fit all $\binom{p}{k}$ models containing exactly k predictors.
 - b. Select the best model among these models and call it $\mathcal{M}_k$. Here, “best” can be defined as the model with the smallest RSS, or equivalently the largest R^2 .
 3. Select a single best model from $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$ using cross-validation, C_p , AIC, BIC, or adjusted R^2 .
- Step 2 identifies the best model for each subset size. Why can we use R^2 here?
 - In Step 3, can we use RSS or R^2 ?

Computational Cost of Best Subset Selection

Best subset selection requires us to fit

$$\binom{p}{0} + \binom{p}{1} + \binom{p}{2} + \cdots + \binom{p}{p}.$$

Using the binomial identity,

$$\sum_{k=0}^p \binom{p}{k} = 2^p$$

models must be considered.

This becomes computationally expensive very quickly as p increases.

The same general idea can be applied to other models, such as logistic regression, although the criterion used to compare models may differ.

Forward Stepwise Selection

Suppose again that

$$\mathbf{X} = (X_1, X_2, X_3).$$

What sequence of models would we consider using **forward stepwise selection**?

Forward Stepwise Selection

Algorithm: Forward Stepwise Selection

1. Let $\mathcal{M}_0$ denote the null model containing no predictors.
2. For $k = 0, \dots, p - 1$:
 - a. Consider all $p - k$ models that add one additional predictor to $\mathcal{M}_k$.
 - b. Choose the best among these models and call it $\mathcal{M}_{k+1}$. Here, “best” is defined as the model with the smallest RSS or largest R^2 .
3. Select a single best model from $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$ using cross-validation, C_p , AIC, BIC, or adjusted R^2 .

Forward Stepwise Selection

Advantage:

Forward stepwise selection is much more computationally efficient than best subset selection.

At iteration k , we compare $p - k$ models.

Therefore, the total number of models considered is

$$1 + \sum_{k=0}^{p-1} (p - k) = 1 + \frac{p(p+1)}{2}.$$

This is much smaller than 2^p .

Disadvantage:

Forward stepwise selection is a **greedy procedure**.

Once a variable enters the model, it is not reconsidered.

Therefore, it is not guaranteed to identify the best model among all possible subsets.

Example: Credit Card Data

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income, student, limit	rating, income, student, limit

Backward Stepwise Selection

Suppose again that

$$\mathbf{X} = (X_1, X_2, X_3).$$

What sequence of models would we consider using **backward stepwise selection**?

Backward Stepwise Selection

Algorithm: Backward Stepwise Selection

1. Let $\mathcal{M}_p$ denote the full model containing all p predictors.
2. For $k = p, p - 1, \dots, 1$:
 - a. Consider all k models obtained by removing one predictor from $\mathcal{M}_k$.
 - b. Choose the best among these models and call it $\mathcal{M}_{k-1}$. Here, “best” is defined as the model with the smallest RSS or largest R^2 .
3. Select a single best model from $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$ using cross-validation, C_p , AIC, BIC, or adjusted R^2 .

Backward Stepwise Selection

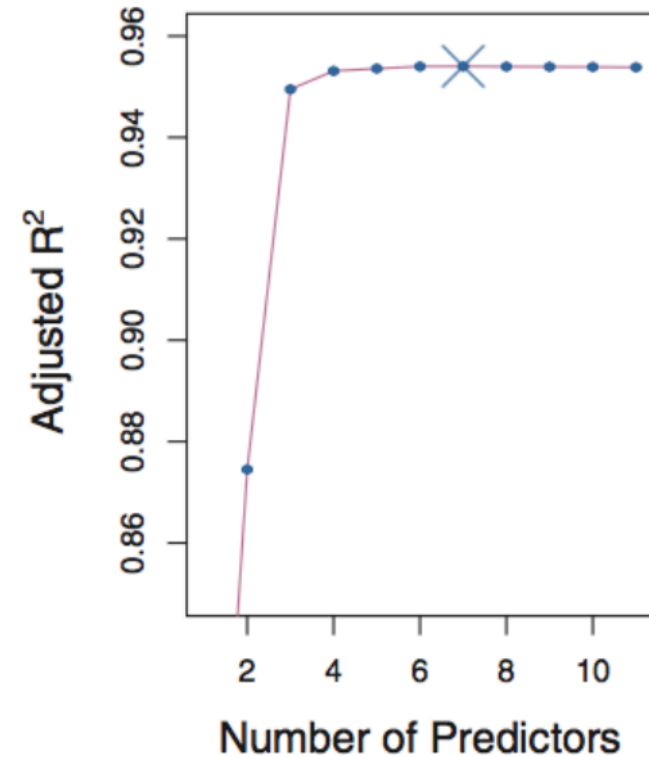
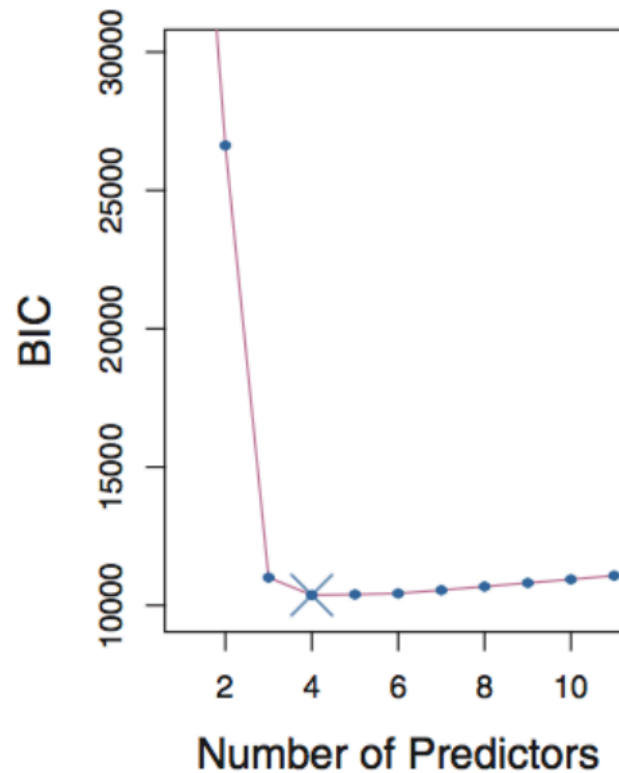
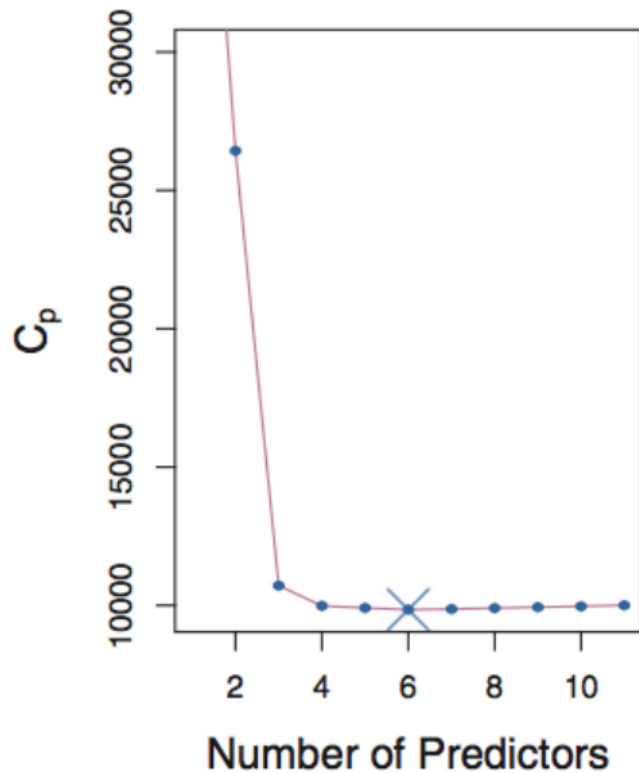
Backward stepwise selection also considers $1 + \frac{p(p+1)}{2}$ models.

This is much smaller than the 2^p models required for best subset selection.

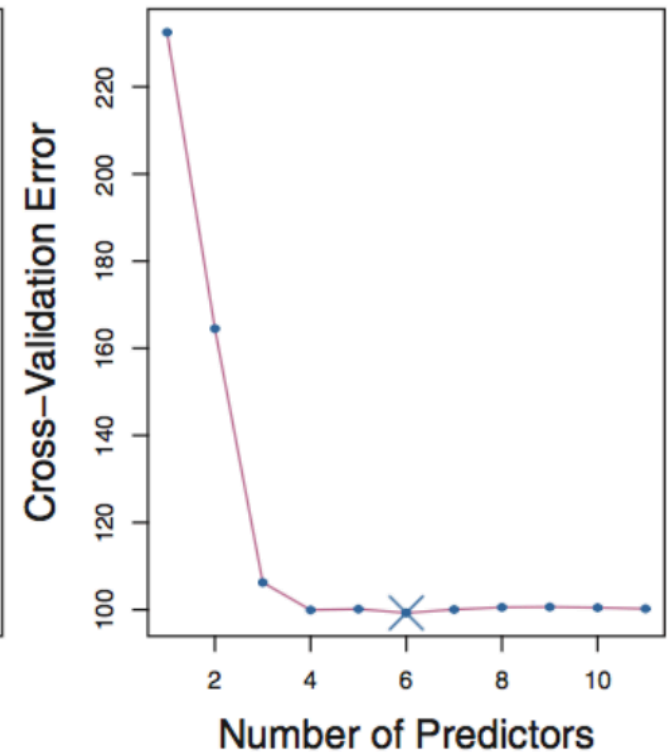
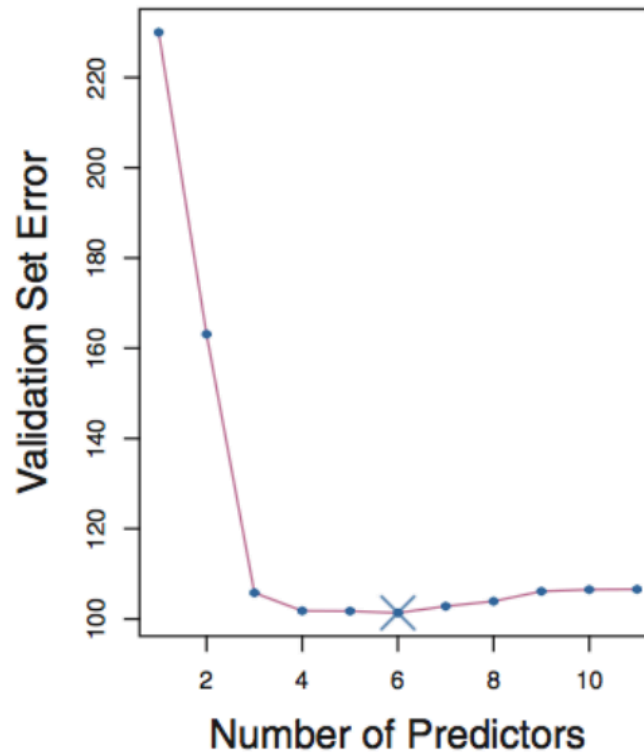
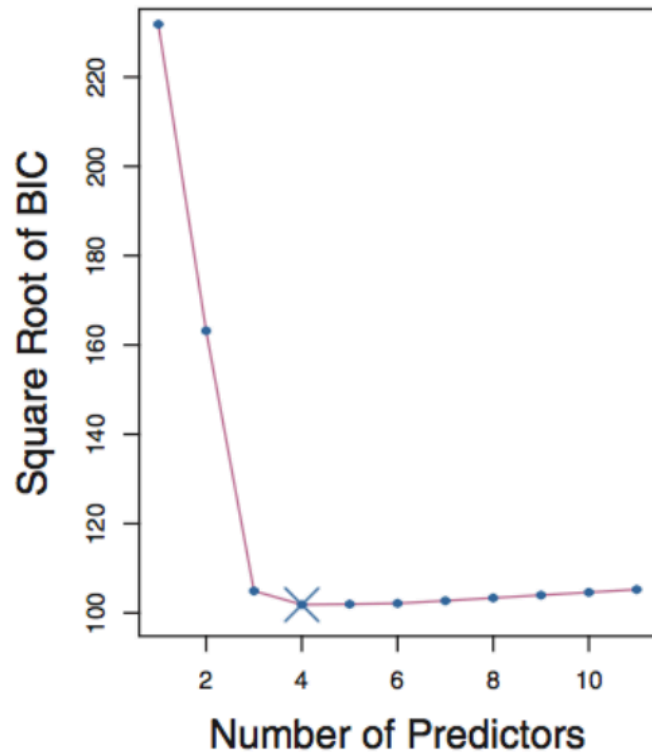
However, backward stepwise selection is also a **greedy procedure**.

It is therefore not guaranteed to identify the globally best subset of predictors.

Credit Card Data: Best Subset Selection via Mallow's C_p , BIC and adjusted R^2



Credit Card Data: Sample Splitting



Summary of This Week

- **Linear regression** is a parametric approach to statistical learning and we can fit the model using **ordinary least squares (OLS)**.
- **Data-splitting methods** can help us estimate test error
 - one single split is unstable
 - can use these methods to compare both parametric and non-parametric models
- **Complexity-adjusted criteria** are tools to help us compare *parametric* models:
 - different criteria optimize different version of the fit-complexity trade-off
- We introduced **subset selection** for choosing which predictors to include in the model
 - best subset selection is computationally expensive, so forward/backward selection, although greedy, are much more realistic

Next Week: Shrinkage + Regularization

Announcements: Tutorials will begin on September 21st, check ACORN for location

